SGN – Assignment #1

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Contents

1 Periodic orbit **2**
1.1 Lagrange points . . . 2
1.2 Halo orbit . . . 4
1.3 Family of Halo orbits . . . 6

2 Impulsive guidance **7**
2.1 Analysis of the problem . . . 8
2.2 Simple shooting . . . 9
2.2.1 Statement of the problem . . . 9
2.2.2 Analytical derivatives . . . 10
2.2.3 Solution of the problem with finite differences gradients . . . 11
2.2.4 Solution of the problem with analytical gradients . . . 12
2.3 Multiple shooting . . . 12
2.3.1 Statement of the problem . . . 12
2.3.2 Analytical derivatives . . . 13
2.3.3 Solution of the problem . . . 15
2.4 Solutions comparison . . . 15
2.5 N-body propagation . . . 16
2.5.1 Identification of the initial epoch . . . 16
2.5.2 Propagation . . . 17

3 Continuous guidance **18**
3.1 Analysis of the problem . . . 19
3.1.1 Debris spatial density . . . 19
3.1.2 Initial state and final state . . . 19
3.2 Adimensionalizzation . . . 20
3.3 Formulation of the problem . . . 20
3.4 Solution for $T_{max} = 3.000$N . . . 22
3.5 Solution for $T_{max} = 2.860$N . . . 24

1 Periodic orbit

Exercise 1

Consider the 3D Earth–Moon Circular Restricted Three-Body Problem with $\mu = 0.012150$. Note that the CRTBP has an integral of motion, that is, the Jacobi constant

$$J(x, y, z, v_x, x_y, v_z) \coloneqq 2\Omega(x, y, z) - v^2 = C$$

where $\Omega(x, y, z) = \frac{1}{2}(x^2 + y^2) + \frac{1-\mu}{r_1} + \frac{\mu}{r_2} + \frac{1}{2}\mu(1-\mu)$ and $v^2 = v_x^2 + v_y^2 + v_z^2$.

1) Find the coordinates of the five Lagrange points L_i in the rotating, adimensional reference frame with at least 10-digit accuracy and report their Jacobi constant C_i.

Solutions to the 3D CRTBP satisfy the symmetry

$$\mathcal{S} : (x, y, z, \dot{x}, \dot{y}, \dot{z}, t) \to (x, -y, z, -\dot{x}, \dot{y}, -\dot{z}, -t).$$

Thus, a trajectory that crosses perpendicularly the $y = 0$ plane twice is a periodic orbit.

2) Given the initial guess $\mathbf{x}_0 = (x_0, y_0, z_0, v_{x0}, v_{y0}, v_{z0})$, with

$$\begin{aligned} x_0 &= 1.068792441776 \\ y_0 &= 0 \\ z_0 &= 0.071093328515 \\ v_{x0} &= 0 \\ v_{y0} &= 0.319422926485 \\ v_{z0} &= 0 \end{aligned}$$

Find the periodic halo orbit having a Jacobi Constant $C = 3.09$; that is, develop the theoretical framework and implement a differential correction scheme that uses the STM, either approximated through finite differences **or** achieved by integrating the variational equation.
Hint: Consider working on $\varphi(\mathbf{x} + \Delta\mathbf{x}, t + \Delta t)$ and $J(\mathbf{x} + \Delta\mathbf{x})$ and then enforce perpendicular cross of $y = 0$ and Jacobi energy.

The periodic orbits in the CRTBP exist in families. These can be computed by 'continuing' the orbits along one coordinate or one parameter, e.g., the Jacobi energy C. The *numerical continuation* is an iterative process in which the desired variable is *gradually* varied, while the rest of the initial guess is taken from the solution of the previous iteration, thus aiding the convergence process.

3) By gradually decreasing C and using numerical continuation, compute the families of halo orbits until $C = 3.04$.

(8 points)

1.1 Lagrange points

The motion of a point in the 3D CRTBP, defined in Equation 1, has 5 equilibrium points, called Lagrange points. The first 3 points are called 'Collinear' and lie on the x-axis, while the last 2 are called 'Triangular' and lie at the vertices of two equilateral triangles with a common base extending from the Earth to the Moon.

$$\begin{cases} \dot{v_x} - 2v_y = \dfrac{\partial \Omega}{\partial x} \\ \dot{v_y} + 2v_x = \dfrac{\partial \Omega}{\partial y} \\ \dot{v_z} = \dfrac{\partial \Omega}{\partial z} \end{cases} \tag{1}$$

The coordinates of L_1, L_2, and L_3 have been found by imposing $y = 0$ and $z = 0$, and then the x coordinate has been found by imposing Equation 2 equal to zero.

$$\frac{\partial \Omega}{\partial x} = x - \frac{1-\mu}{r_1^3}(x-\mu) - \frac{\mu}{r_2^3}(x+\mu-1) \tag{2}$$

The coordinates of L_4, and L_5 have been found by imposing $x = \frac{1}{2} - mu$ and $z = 0$, and then y coordinate has been found by imposing Equation 3 equal to zero.

$$\frac{\partial \Omega}{\partial y} = y - \frac{1-\mu}{r_1^3}y - \frac{\mu}{r_2^3}y \tag{3}$$

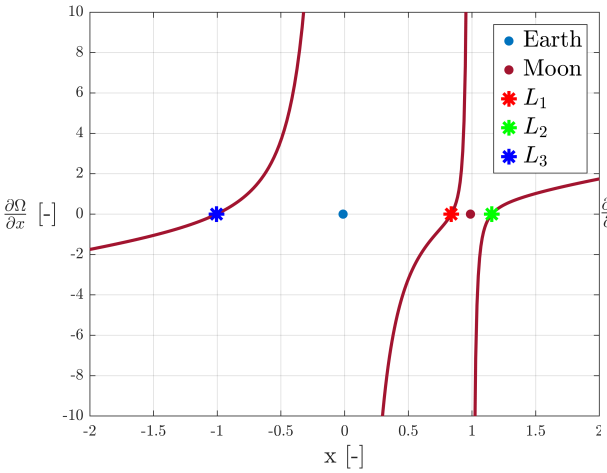


Figure 1: Lagrange points L_1, L_2, and L_3

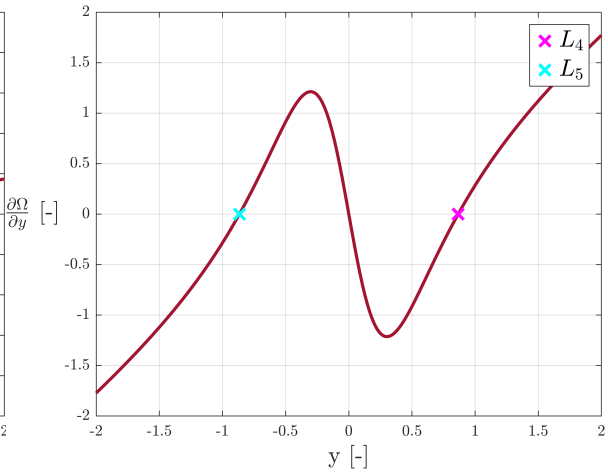


Figure 2: Lagrange points L_4, and L_5

The solution has been found using MATLAB `fzero` function with carefulness in choosing the initial guess cause it is fundamental for finding the solution.
Once all the coordinates have been found, it is possible to compute the associated Jacobi constant (C) by plugging the coordinates into J. The coordinates and the C associated with the Lagrange points are reported in Table 1.

	L_1	L_2	L_3	L_4	L_5
x	0.8369180073	1.1556799131	−1.0050624018	0.4878500000	0.4878500000
y	0.0000000000	0.0000000000	0.0000000000	0.8660254038	−0.8660254038
C	3.2003380950	3.1841582164	3.0241489429	3.0000000000	3.0000000000

Table 1: Lagrangian points coordinates and Jacobi constants

An important check to verify the validity of the results involves confirming the condition: $C_1 > C_2 > C_3 > C_4 = C_5$ which, as shown in Table 1, is satisfied.

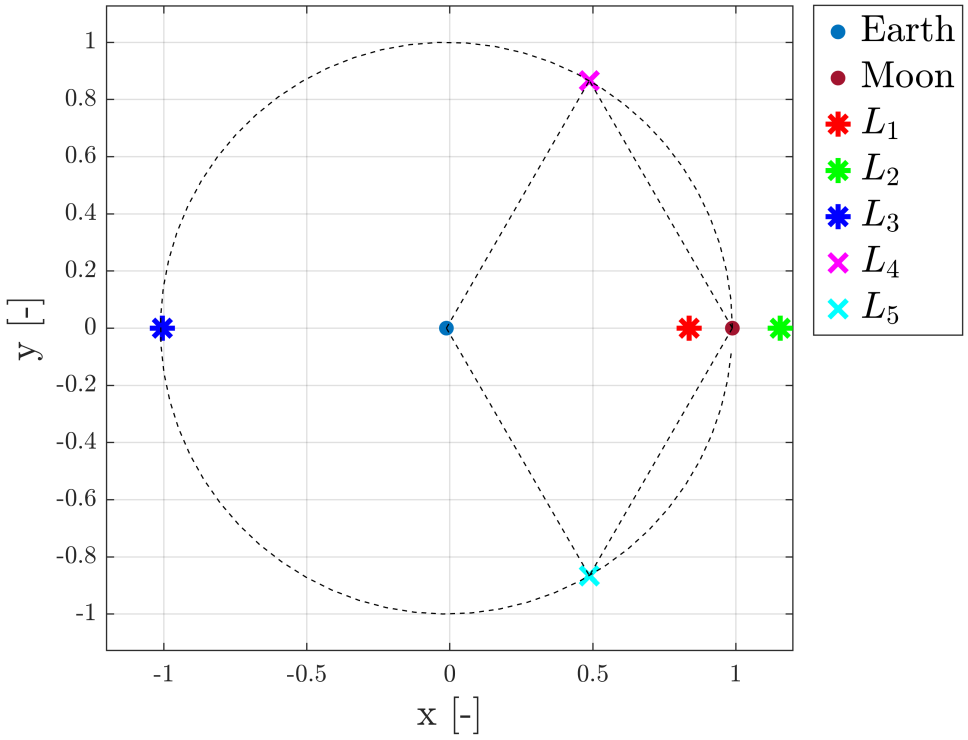


Figure 3: Location of the 5 Lagrange points in the Earth-Moon system

1.2 Halo orbit

A halo orbit is a periodic orbit around a collinear Lagrangian point. To achieve a successful periodic motion with a specific value of C, adjustments to the initial guess need to be made. These corrections are made involving the variational approach through the State Transition Matrix (STM).
The Equation 4 represents the statement of the problem, it consists of 42 odes, that need to be solved at every time step.

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t) & \mathbf{x}(t_0) = \mathbf{x}_0 \\ \dot{\mathbf{\Phi}}(t_0, t) = \left[\dfrac{\partial \mathbf{f}}{\partial \mathbf{x}}\right] \mathbf{\Phi}(t_0, t) & \mathbf{\Phi}(t_0, t_0) = \mathbf{I}_{6x6} \end{cases} \tag{4}$$

The propagation is performed on a time interval $[t_0, t_f]$, then the state and STM at the final time are employed in the differential correction scheme reported in Equation 5 and Equation 6. The subscript f stands for the desired value at the final time.

$$\varphi(\mathbf{x_0} + \Delta\mathbf{x_0}, t_0 + \Delta t_0) = \varphi(\mathbf{x_0}, t_0) + \mathbf{\Phi}(\mathbf{x}_0, t_0, t_0 + \Delta t_0)\Delta\mathbf{x_0} + \mathbf{f}(\mathbf{x}_0, t_0)\Delta t_0 = \begin{bmatrix} x_f \\ 0 \\ z_f \\ 0 \\ v_{y,f} \\ 0 \end{bmatrix} \tag{5}$$

$$J(\mathbf{x_0} + \Delta\mathbf{x_0}) = J(\mathbf{x_0}) + \left.\frac{\partial J}{\partial \mathbf{x_0}}\right|_{\mathbf{x_0}} \Delta\mathbf{x_0} = C_f \tag{6}$$

By manipulating Equation 5 and Equation 6 and isolating the lines 2, 4, and 6, the system in Equation 7 is obtained. Iteratively, the system has been solved and the corrections are applied to the initial state and initial time. The procedure repeats until all y_f^e, $v_{x,f}^e$ and $v_{z,f}^e$ approaches

zero and the same time C_f^e approaches the desired value, which in this case is 3.09.

$$\begin{bmatrix} -y_f^e \\ -v_{x,f}^e \\ -v_{z,f}^e \\ C_f - C_f^e \end{bmatrix} = \begin{bmatrix} \Phi_{2,1} & \Phi_{2,3} & \Phi_{2,5} & \mathbf{f}_2(\mathbf{x_0}, t_0) \\ \Phi_{4,1} & \Phi_{4,3} & \Phi_{4,5} & \mathbf{f}_4(\mathbf{x_0}, t_0) \\ \Phi_{6,1} & \Phi_{6,3} & \Phi_{6,5} & \mathbf{f}_6(\mathbf{x_0}, t_0) \\ \left.\frac{\partial J}{\partial x}\right|_{\mathbf{x_0}} & \left.\frac{\partial J}{\partial z}\right|_{\mathbf{x_0}} & \left.\frac{\partial J}{\partial v_y}\right|_{\mathbf{x_0}} & 0 \end{bmatrix} \begin{bmatrix} \delta x_0 \\ \delta z_0 \\ \delta v_{y0} \\ \delta t_0 \end{bmatrix} \tag{7}$$

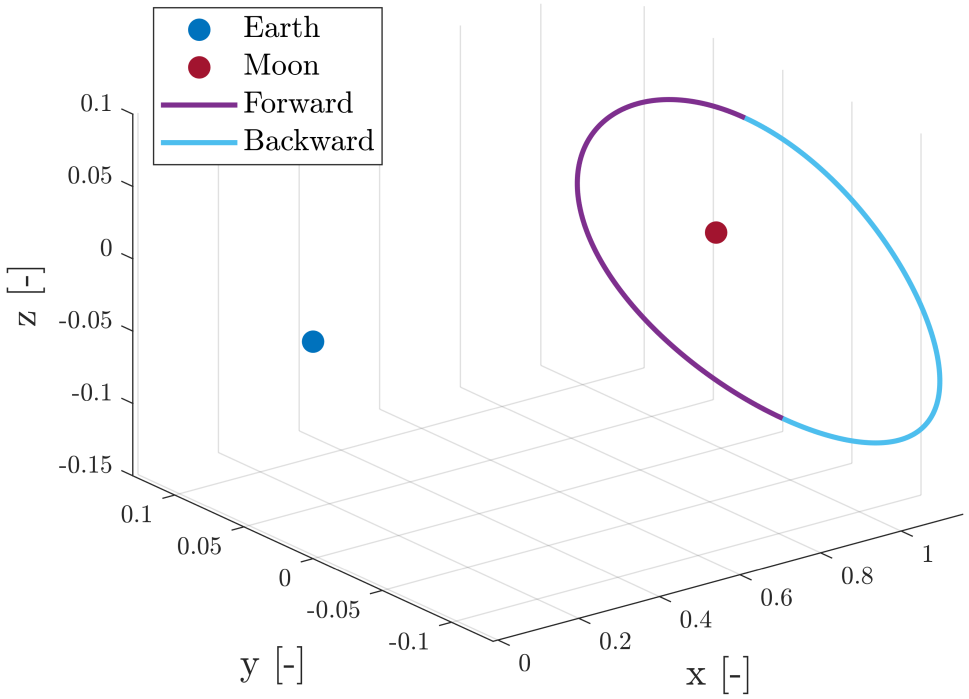


Figure 4: Halo orbit with $C = 3.09$
@Earth-Moon Barycenter ECLIPJ2000 Adimensional rotating reference frame

x	y	z
1.0590402077	0.0000000000	0.0739277378

v_x	v_x	v_x
0.0000000000	0.3469245709	0.0000000000

Table 2: Corrected initial state of the halo orbit with $C = 3.09$

T
3.2157469063

Table 3: Period of the halo orbit with $C = 3.09$

1.3 Family of Halo orbits

The procedure that has been implemented for computing the family of halo orbits consists of, using the differential correction explained in subsection 1.2 with numerical continuation.
The algorithm is the following:

- decreasing the value of the C_f;
- performing the differential correction;
- using as the initial state for the next iteration the state one computed before.

The state used for the first iteration is the one with $C = 3.09$, then this procedure is repeated until C reaches 3.04.

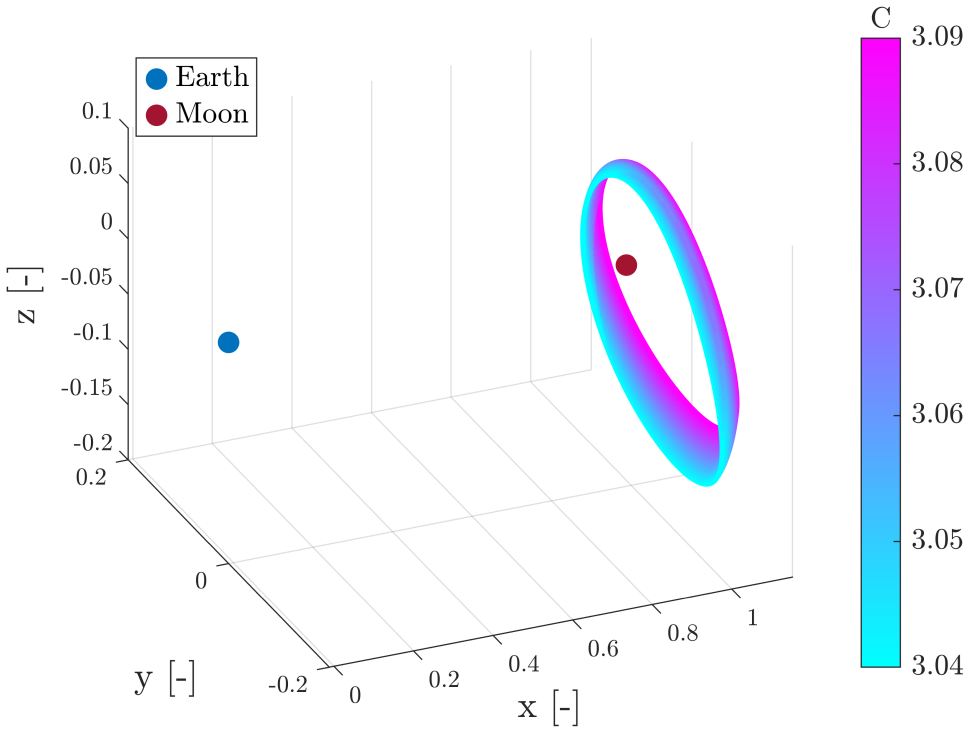


Figure 5: Family of 20 halo orbits with $C \in [3.04, 3.09]$
@Earth-Moon Barycenter ECLIPJ2000 Adimensional rotating reference frame

x	y	z
1.0125655235	0.0000000000	0.0672339583

v_x	v_y	v_x
0.0000000000	0.5103251959	0.0000000000

Table 4: Corrected initial state of the halo orbit with $C = 3.04$

T
2.8444344182

Table 5: Period of the halo orbit with $C = 3.04$

2 Impulsive guidance

Exercise 2

Consider the two-impulse transfer problem stated in Section 3.1 (Topputo, 2013)*.

1) Using the procedure in Section 3.2, produce a first guess solution using $\alpha = 0.2\pi$, $\beta = 1.41$, $\delta = 4$, and $t_i = 2$. Plot the solution in both the rotating frame and Earth-centered inertial frame (see Appendix 1 in (Topputo, 2013)). Consider the parameters listed in Table 6 and extract the radius and gravitational parameters of the Earth and Moon from the provided kernels and use the latter to compute the parameter μ.

Symbol	Value	Units	Meaning
m_s	3.28900541×10^5	-	Scaled mass of the Sun
ρ	3.88811143×10^2	-	Scaled Sun-(Earth+Moon) distance
ω_s	$-9.25195985 \times 10^{-1}$	-	Scaled angular velocity of the Sun
ω_{em}	$2.66186135 \times 10^{-1}$	s^{-1}	Earth-Moon angular velocity
l_{em}	3.84405×10^8	m	Earth-Moon distance
h_i	167	km	Altitude of departure orbit
h_f	100	km	Altitude of arrival orbit
DU	3.84405000×10^5	km	Distance Unit
TU	4.34256461	days	Time Unit
VU	1.02454018	km/s	Velocity Unit

Table 6: Constants to be considered to solve the PBRFBP. The units of distance, time, and velocity are used to map scaled quantities into physical units.

2) Considering the first guess in 1) and using $\{\mathbf{x}_i, t_i, t_f\}$ as variables, solve the problem in Section 3.1 with simple shooting in the following cases

   a) without providing any derivative to the solver, and

   b) by providing the derivatives and by estimating the state transition matrix with variational equations.

3) Considering the first guess solution in 1) and the procedure in Section 3.3, solve the problem with multiple shooting taking $N = 4$ and using the variational equation to compute the Jacobian of the nonlinear equality constraints.

4) Perform an n-body propagation using the solution $\{\mathbf{x}_i, t_i, t_f\}$ obtained in point 2), transformed in Earth-centered inertial frame and into physical units. To move from 2-D to 3-D, assume that the position and velocity components in inertial frame are $r_z(t_i) = 0$ and $v_z(t_i) = 0$. To perform the propagation it is necessary to identify the epoch t_i. This can be done by mapping the relative position of the Earth, Moon and Sun in the PCRTBP to a similar condition in the real world:

   a) Consider the definition of $\theta(t)$ provided in Section 2.2 to compute the angle $\theta_i = \theta(t_i)$. Note that this angle corresponds to the angle between the rotating frame x-axis, aligned to the position vector from the Earth-Moon System Barycenter (EMB) to the Moon, and the Sun direction.

   b) The angle θ ranges between $[0, 2\pi]$ and it covers this domain in approximately the revolution period of the Moon around the Earth.

*F. Topputo, "On optimal two-impulse Earth–Moon transfers in a four-body model", Celestial Mechanics and Dynamical Astronomy, Vol. 117, pp. 279–313, 2013, DOI: 10.1007/s10569-013-9513-8.

c) Solve a zero-finding problem to determine the epoch at which the angle Moon-EMB-Sun is equal to θ_i, considering as starting epoch `2024 Sep 28 00:00:00.000 TDB`. **Hints**: Exploit the SPK kernels to define the orientation of the rotating frame axes in the inertial frame for an epoch t. Consider only the projection of the EMB-Sun position vector onto the so-defined x-y plane to compute the angle (planar motion).

Plot the propagated orbit and compare it to the previously found solutions.

(11 points)

2.1 Analysis of the problem

Before proceeding into the computation of the initial guess, it is necessary to compute first the Earth–Moon mass parameter (μ), which has been computed as:

$$\mu = \frac{\mu_E}{\mu_E + \mu_M} \tag{8}$$

The value of both gravitational parameters, the one of the Earth and the one of the Moon have been extracted from the kernels using `cspice_bodvrd`. The value of μ obtained is the following:

μ
0.0121505843

Table 7: Earth–Moon mass parameter

Then the initial guess reported in Table 8, has been computed using Equation 18 in Section 3.2 of (Topputo, 2013)[†]. The equations of motions of the problem are represented in Equation 9,

$r_{x,0}$ $[DU]$	$r_{y,0}$ $[DU]$	$v_{x,0}$ $[VU]$	$v_{y,0}$ $[VU]$
0.001624	0.010008	-6.302738	8.674975

Table 8: Initial guess in Earth-Moon rotating frame.

where $\mathbf{f}$ is the Equation 5 in Section 2.2 of (Topputo, 2013)†, and $\mathbf{\Phi}$ is the STM: the solver computes results of 20 scalar odes per step.

$$\begin{cases} \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t) & \mathbf{x}(t_0) = \mathbf{x}_0 \\ \dot{\mathbf{\Phi}}(t_0, t) = \left[\dfrac{\partial \mathbf{f}}{\partial x}\right] \mathbf{\Phi}(t_0, t) & \mathbf{\Phi}(t_0, t_0) = \mathbf{I}_{4x4} \end{cases} \tag{9}$$

By propagating the initial guess it is possible to observe that the trajectory does not arrive to the Moon, as shown in Figure 6, and Figure 7.

[†]F. Topputo, "On optimal two-impulse Earth–Moon transfers in a four-body model", Celestial Mechanics and Dynamical Astronomy, Vol. 117, pp. 279–313, 2013, DOI: 10.1007/s10569-013-9513-8.

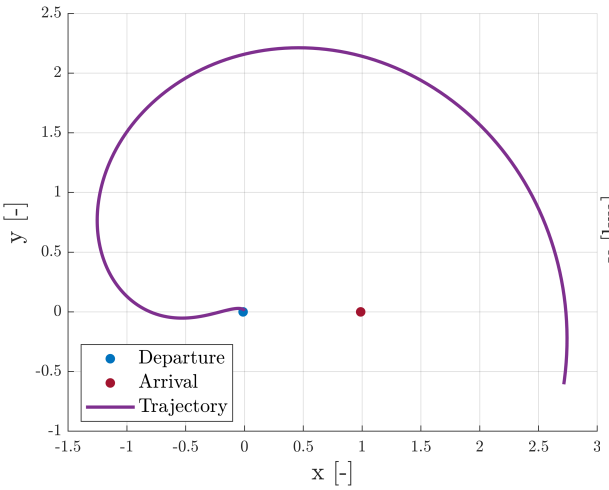


Figure 6: Initial guess propagation
@Earth-Moon Barycenter
Earth-Moon Adimensional rotating frame

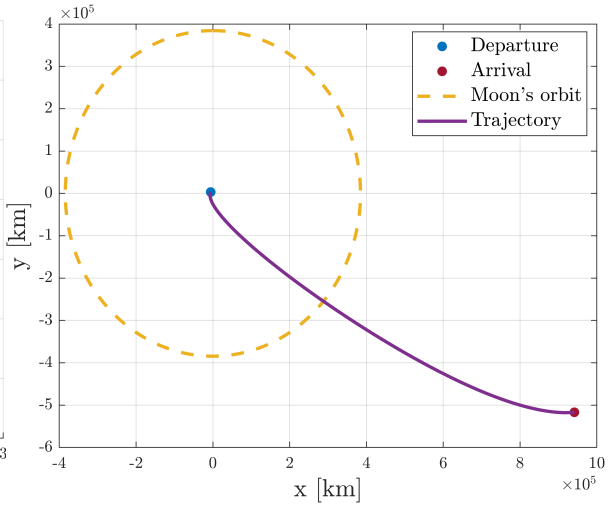


Figure 7: Initial guess propagation
@Earth
Earth Centered Inertial Frame

2.2 Simple shooting

2.2.1 Statement of the problem

The overall problem can be conceptualized as an algebraic Non Linear Programming (NLP) problem. The object is to minimize an object function while the equality and inequality constraints are satisfied. In the simple shooting approach, an initial state $\mathbf{x}_i$, in the Earth-Moon adimensional rotating frame is propagated from t_i to t_f, then the objective function and the constraints are computed. The variables of the problem are the initial state, the initial and final time, for a total of 6 variables:

$$\mathbf{y} = [\mathbf{x}_i,\, t_i,\, t_f] \tag{10}$$

The problem for the simple shooting is stated as follows:

Let $\mathbf{x}(t) = \varphi(\mathbf{x_i}, t_i, t)$ be a solution of Equation 9, the optimization for the two-impulse Earth-Moon transfer consists in:

$$\min_{\mathbf{y}:=(\mathbf{x_i},t_i,t_f)} F_{ss}(\mathbf{y}) \quad s.t. \begin{cases} \mathbf{C}_{eq,ss}(\mathbf{y}) = 0 \\ \mathbf{C}_{ss}(\mathbf{y}) < 0 \end{cases} \tag{11}$$

Where $F_{ss}(\mathbf{y})$ is the objective function and $\mathbf{C}_{eq,ss}(\mathbf{y})$, $\mathbf{C}_{ss}(\mathbf{y})$ are the equality and inequality constraints respectively.

Objective function

$$F_{ss}(\mathbf{y}) = \Delta v(\mathbf{x}_i, t_i, t_f) = \Delta v_i(\mathbf{x}_i) + \Delta v_f(\varphi(\mathbf{x}_i, t_i, t_f)) \tag{12}$$

$$\Delta v_i = v_i - \sqrt{\frac{(1-\mu)}{r_i}} \qquad v_i = \sqrt{(v_{x,i} - y_i)^2 + (v_{y,i} + x_i + \mu)^2}$$

$$\Delta v_f = v_f - \sqrt{\frac{\mu}{r_f}} \qquad v_f = \sqrt{(v_{x,f} - y_f)^2 + (v_{y,f} + x_f + \mu - 1)^2}$$

Equality constraints

$$\mathbf{C}_{eq,ss}(\mathbf{y}) = \begin{bmatrix} \boldsymbol{\psi}_i \\ \boldsymbol{\psi}_f \end{bmatrix} \tag{13}$$

$$\boldsymbol{\psi}_i(\mathbf{x}_i) = \begin{bmatrix} (x_i+\mu)^2 + y_i^2 - r_i^2 \\ (x_i+\mu)(v_{x,i}-y_i) + y_i(v_{y,i}+x_i+mu) \end{bmatrix} \tag{14}$$

$$\boldsymbol{\psi}_f(\mathbf{x}_f) = \begin{bmatrix} (x_f+\mu-1)^2 + y_f^2 - r_f^2 \\ (x_f+\mu-1)(v_{x,f}-y_f) + y_f(v_{y,f}+x_f+mu-1) \end{bmatrix} \tag{15}$$

Inequality constraints

$$C_{ss}(\mathbf{y}) = \tau = \begin{bmatrix} t_i - t_f \end{bmatrix} \tag{16}$$

2.2.2 Analytical derivatives

Before proceeding with the computation of the analytical derivatives, is fundamental to recall the derivative of the flow ($\varphi(\mathbf{x_i}, t_i, t_f)$) with respect to $\mathbf{x_i}$, t_i, and t_f.

$$\frac{\partial \varphi(\mathbf{x_0}, t_0, t_f)}{\partial \mathbf{x_i}} = \boldsymbol{\Phi}(t_i, t_f) \tag{17}$$

$$\frac{\partial \varphi(\mathbf{x_i}, t_i, t_f)}{\partial t_i} = -\boldsymbol{\Phi}(t_i, t_f)\mathbf{f}(\mathbf{x_i}, t_i), \qquad \frac{\partial \varphi(\mathbf{x_i}, t_i, t_f)}{\partial t_f} = \mathbf{f}(\mathbf{x}(t_f), t_f), \tag{18}$$

Objective function

In the following expression is reported the analytical gradient of the objective function with respect to the variable vector $\mathbf{y}$.

$$\frac{\partial F_{ss}}{\partial \mathbf{y}} = \begin{bmatrix} v_{i/\mathbf{y}} + v_{f/\mathbf{y}} \end{bmatrix} \tag{19}$$

$$\mathbf{v}_{i/\mathbf{y}} = \begin{bmatrix} \frac{(v_{y,i}+x_i+\mu)}{v_i} & -\frac{(v_{x,i}-y_i)}{v_i} & \frac{(v_{x,i}-y_i)}{v_i} & \frac{(v_{y,i}+x_i+\mu)}{v_i} & 0 & 0 \end{bmatrix} \tag{20}$$

$$\boldsymbol{v}_{f/\mathbf{y}} = \boldsymbol{v}_{f/\mathbf{x}_f} \begin{bmatrix} \boldsymbol{\Phi}(t_0, t_f) & \mathbf{0}_{2x2} \\ \mathbf{0}_{2x4} & \mathbf{F}_{/t} \end{bmatrix}^T \tag{21}$$

$$\boldsymbol{v}_{f/\mathbf{x}_f} = \begin{bmatrix} \frac{(v_{y,f}+x_f+\mu-1)}{v_f} & -\frac{(v_{x,f}-y_f)}{v_f} & \frac{(v_{x,f}-y_f)}{v_f} & \frac{(v_{y,f}+x_f+\mu-1)}{v_f} & \frac{(v_{x,f}-y_f)}{v_f} & \frac{(v_{y,f}+x_f+\mu-1)}{v_f} \end{bmatrix} \tag{22}$$

$$\mathbf{F}_{/t} = \begin{bmatrix} (\varphi_{/t_i})_3 - (\varphi_{/t_i})_2 & (\varphi_{/t_i})_4 + (\varphi_{/t_i})_1 \\ (\mathbf{f}_3(\mathbf{x}_f, t_f)) - (\mathbf{f}_2(\mathbf{x}_f, t_f)) & (\mathbf{f}_4(\mathbf{x}_f, t_f)) + (\mathbf{f}_1(\mathbf{x}_f, t_f)) \end{bmatrix} \tag{23}$$

Where ($\varphi_{/t_i}$) corresponds to Equation 18.

Equality constraints

In the following are reported the derivatives of the equality constraints with respect to the variable vector $\mathbf{y}$.

$$\frac{\partial \mathbf{C}_{eq,ss}}{\partial \mathbf{y}} = \begin{bmatrix} 2(x_i+\mu) & 2y_i & 0 & 0 & 0 & 0 \\ v_{x,i} & v_{y,i} & (x_i+\mu) & y_i & 0 & 0 \\ & & \boldsymbol{\psi}_{(f,1)/\mathbf{y}} & & & \\ & & \boldsymbol{\psi}_{(f,2)/\mathbf{y}} & & & \end{bmatrix} \tag{24}$$

$$\boldsymbol{\psi}_{(f,1)/\mathbf{y}} = \begin{bmatrix} 2(x_f+\mu-1) & 2y_f & 0 & 0 & 2(x_f+\mu-1) & 2y_f \end{bmatrix} \begin{bmatrix} \boldsymbol{\Phi}(t_0,t_f) & \mathbf{0}_{2x2} \\ \mathbf{0}_{2x4} & \mathbf{C}_{3/t} \end{bmatrix}^T \tag{25}$$

$$\boldsymbol{\psi}_{(f,2)/\mathbf{y}} = \begin{bmatrix} v_{x,f} & v_{y,f} & (x_f+\mu-1) & y_f \end{bmatrix} \begin{bmatrix} \boldsymbol{\Phi}(t_0,t_f) \\ (-\boldsymbol{\Phi}(\mathbf{t_0},\mathbf{t_f})\mathbf{f}(\mathbf{x}_i,t_i))^T \\ \mathbf{f}^T(\mathbf{x}_f,t_f) \end{bmatrix}^T \tag{26}$$

$$\mathbf{C}_{3/t} = \begin{bmatrix} (\varphi_{/t_i})_1 & (\varphi_{/t_i})_2 \\ \mathbf{f}(\mathbf{x}_i,t_i))_1 & \mathbf{f}(\mathbf{x}_i,t_i))_2 \end{bmatrix} \tag{27}$$

Where $(\varphi_{/t_i})$ corresponds to Equation 18.

Inequality constraints

$$\frac{\partial C_{ss}}{\partial \mathbf{y}} = \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & -1 \end{bmatrix} \tag{28}$$

2.2.3 Solution of the problem with finite differences gradients

The problem has been solved through MATLAB solver `fmincon`, using the initial guess reported in Table 8. Given the specified objective function and constraints the solver determines the optimal value of the optimization variables vector $\mathbf{y}$ which minimizes the value of the objective function. In this case, the derivatives are computed using the central finite differences.

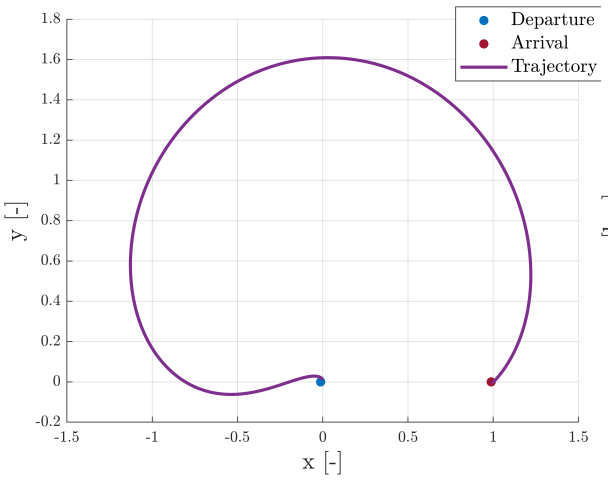


Figure 8: Earth-Moon transfer simple shooting without gradients
@Earth-Moon Barycenter
Earth-Moon Adimensional rotating frame

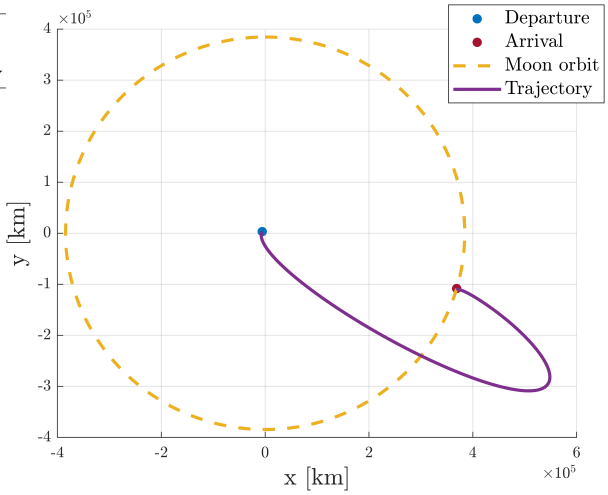


Figure 9: Earth-Moon transfer simple shooting without gradients
@Earth
Earth Centered Inertial Frame

2.2.4 Solution of the problem with analytical gradients

The problem has been solved using the same procedure explained in subsubsection 2.2.3, but in this case, the analytic gradients of the objective function, equality, and inequality, have been provided to the solver. The advantage of providing them consists of fast and accurate computations of optimal solutions.

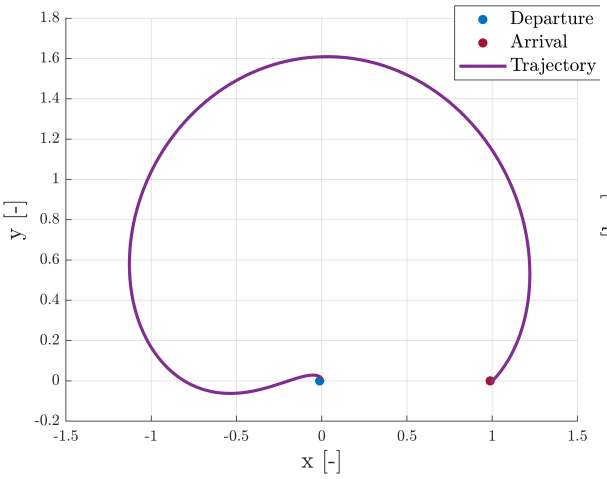


Figure 10: Earth-Moon transfer simple shooting with gradients
@Earth-Moon Barycenter
Earth-Moon Adimensional rotating frame

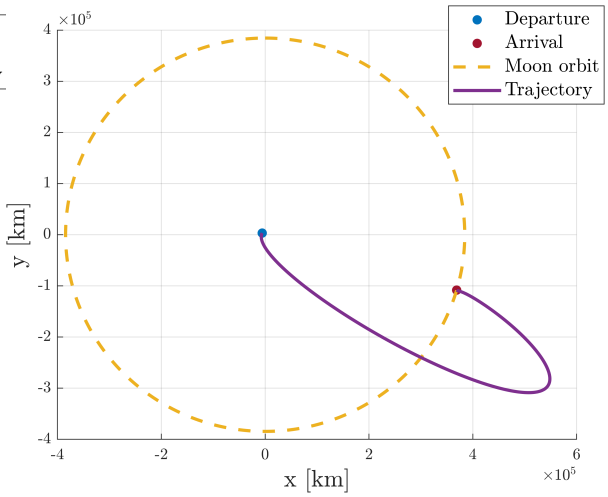


Figure 11: Earth-Moon transfer simple shooting with gradients
@Earth
Earth Centered Inertial Frame

Gradients	$r_{x,0}$ $[DU]$	$r_{y,0}$ $[DU]$	$v_{x,0}$ $[VU]$	$v_{y,0}$ $[VU]$	t_i $[TU]$	t_f $[TU]$
False	0.001624	0.010009	-6.288004	8.654056	1.998	6.003
True	0.001624	0.010009	-6.288003	8.654058	1.998	6.003

Table 9: Simple shooting solutions in the Earth-Moon rotating frame.

2.3 Multiple shooting

2.3.1 Statement of the problem

As for the single shooting problem also the problem of multiple shooting has been conceptualized as an algebraic NLP. A first guess solution has been propagated from t_i to t_f then the trajectory has been divided into 3, segments which correspond to 4 nodes equally spaced in time:

$$t_j = t_i + \frac{j-1}{N-1}(t_f - t_i), \qquad j = 1, ..., 4 \tag{29}$$

Then, each $j-th$ mesh point, from $j = 1, ..., N$. The multiple shooting method requires the dynamics Equation 9 to be integrated within the $N-1$ intervals $[t_j, t_{j+1}]$ by taking $\mathbf{x}_j$ as initial condition. The variables of the problem are the 4 states, the initial and final time, for a total of 18 variables:

$$\mathbf{y} = [\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \mathbf{x}_4, t_i, t_f] \tag{30}$$

The problem for the multiple shooting is stated as follows:

Let $\mathbf{x}(t) = \varphi(\mathbf{x}_j, t_j, t_{j+1})$ *be a solution of Equation 9 in* $[t_j, t_{j+1}]$ *from* $j = 1, ..., N$*, the optimization for the two-impulse Earth-Moon transfer consists in:*

$$\min_{\mathbf{y}:=(\mathbf{x_1},\mathbf{x_2},\mathbf{x_3},\mathbf{x_4},t_i,t_f)} F_{ms}(\mathbf{y}) \quad s.t. \begin{cases} \mathbf{C}_{eq,ms}(\mathbf{y}) = 0 \\ \mathbf{C}_{ms}(\mathbf{y}) < 0 \end{cases} \tag{31}$$

Where $F_{ms}(\mathbf{y})$ is the objective function and $\mathbf{C}_{eq,ms}(\mathbf{y})$, $\mathbf{C}_{ms}(\mathbf{y})$ are the equality and inequality constraints respectively.

Objective function

The objective function changes a bit with respect to the simple shooting one. The Δv_i is the same, while the Δv_f instead of being evaluated using the flow is computed using the last state of the variable vector.

$$F_{ms}(\mathbf{y}) = \Delta v(\mathbf{x}_i, t_i, t_f) = \Delta v_i(\mathbf{x}_i) + \Delta v_f(\mathbf{x}_4) \tag{32}$$

$$\Delta v_i = v_1 - \sqrt{\frac{(1-\mu)}{r_i}} \qquad v_1 = \sqrt{(v_{x,1} - y_1)^2 + (v_{y,1} + x_1 + \mu)^2}$$

$$\Delta v_f = v_f - \sqrt{\frac{\mu}{r_f}} \qquad v_f = \sqrt{(v_{x,4} - y_4)^2 + (v_{y,4} + x_4 + \mu - 1)^2}$$

Equality constraints

The equality constraints of the simple shooting are maintained and have been added to the constraints that grant the continuity of position and velocity. For the single mesh's point are necessary 4 constraints, so a total of $4(N-1) = 12$, because the continuity needs are integrated within $N-1$ time interval. This leads to a number of equality constraints equal to $4N = 16$.

$$\boldsymbol{\zeta}_j = \varphi(\mathbf{x}_j, t_j, t_{j+1}) - \mathbf{x}_{j+1} \qquad j = 1, ..., 3 \tag{33}$$

$$\mathbf{C}_{eq,ms}(\mathbf{y}) = \begin{bmatrix} \boldsymbol{\zeta}_1 & \boldsymbol{\zeta}_2 & \boldsymbol{\zeta}_3 & \boldsymbol{\psi}_i(\mathbf{x}_1) & \boldsymbol{\psi}_f(\mathbf{x}_4) \end{bmatrix}^T \tag{34}$$

Inequality constraints

The inequality constraints added to the previous one are used to avoid solutions that impact either the Earth or the Moon. For the single mesh point are necessary two constraints one to avoid the impact on the Earth and one to avoid the impact on the Moon. This leads to a number of inequality constraints equal to $2N + 1 = 9$.

$$\boldsymbol{\eta}_j = \begin{bmatrix} (R_e/DU)^2 - (x_j + \mu)^2 - y_j^2 \\ (R_m/DU)^2 - (x_j + \mu - 1)^2 - y_j^2 \end{bmatrix} \qquad j = 1, ..., 4 \tag{35}$$

$$C_{ms}(\mathbf{y}) = \begin{bmatrix} \boldsymbol{\eta}_1 & \boldsymbol{\eta}_2 & \boldsymbol{\eta}_3 & \boldsymbol{\eta}_4 & \tau \end{bmatrix}^T \tag{36}$$

2.3.2 Analytical derivatives

Objective function

The derivative of the objective function for the multiple shooting case is simpler compared to the single shooting one because the derivative of the flow does not appear.

$$\frac{\partial F_{ms}}{\partial \mathbf{y}} = \begin{bmatrix} \boldsymbol{v}_{1/\mathbf{x}_1} & \mathbf{0}_{8x1} & \boldsymbol{v}_{f/\mathbf{x}_4} & \mathbf{0}_{2x1} \end{bmatrix} \tag{37}$$

$$\boldsymbol{v}_{1/\mathbf{x}_1} = \begin{bmatrix} \dfrac{(v_{y,1} + x_1 + \mu)}{v_1} & -\dfrac{(v_{x,1} - y_1)}{v_1} & \dfrac{(v_{x,1} - y_1)}{v_1} & \dfrac{(v_{y,1} + x_1 + \mu)}{v_1} \end{bmatrix} \tag{38}$$

$$\boldsymbol{v}_{f/\mathbf{x}_4} = \begin{bmatrix} \dfrac{(v_{y,4} + x_4 + \mu - 1)}{v_f} & -\dfrac{(v_{x,4} - y_4)}{v_f} & \dfrac{(v_{x,4} - y_4)}{v_f} & \dfrac{(v_{y,4} + x_4 + \mu - 1)}{v_f} \end{bmatrix} \tag{39}$$

Equality constraints

The derivative of the equality constraints vector is a matrix of 16×18, the Equation 41 and Equation 42 are different from zero because as specified in Equation 29 all the time instant are equally spaced so every t_j is dependent from t_i and t_f.

$$\frac{\partial \mathbf{C}_{eq,ms}}{\partial \mathbf{y}} = \begin{bmatrix} \boldsymbol{\Phi}(t_i,\, t_2) & -\mathbf{I}_{4x4} & \mathbf{0}_{4x4} & \mathbf{0}_{4x4} & \boldsymbol{\zeta}_{1/t_i} & \boldsymbol{\zeta}_{1/t_f} \\ \mathbf{0}_{4x4} & \boldsymbol{\Phi}(t_2,\, t_3) & -\mathbf{I}_{4x4} & \mathbf{0}_{4x4} & \boldsymbol{\zeta}_{2/t_i} & \boldsymbol{\zeta}_{2/t_f} \\ \mathbf{0}_{4x4} & \mathbf{0}_{4x4} & \boldsymbol{\Phi}(t_3,\, t_f) & -\mathbf{I}_{4x4} & \boldsymbol{\zeta}_{3/t_i} & \boldsymbol{\zeta}_{3/t_f} \\ \boldsymbol{\psi}_{i/\mathbf{x}_1} & \mathbf{0}_{2x4} & \mathbf{0}_{2x4} & \mathbf{0}_{2x4} & \mathbf{0}_{2x4} & \mathbf{0}_{2x4} \\ \mathbf{0}_{2x4} & \mathbf{0}_{2x4} & \mathbf{0}_{2x4} & \boldsymbol{\psi}_{f/\mathbf{x}_4} & \mathbf{0}_{2x4} & \mathbf{0}_{2x4} \end{bmatrix} \tag{40}$$

$$\boldsymbol{\zeta}_{j/t_i} = \left[-\boldsymbol{\Phi}(t_j, t_{j+1}) \mathbf{f}(\mathbf{x}_j, t_j) \frac{4-j}{3} + \mathbf{f}(\mathbf{x}_{j+1}, t_{j+1}) \frac{3-j}{3} \right] \quad j = 1,\, 2,\, 3 \tag{41}$$

$$\boldsymbol{\zeta}_{j/t_f} = \left[-\boldsymbol{\Phi}(t_j, t_{j+1}) \mathbf{f}(\mathbf{x}_j, t_j) \frac{j-1}{3} + \mathbf{f}(\mathbf{x}_{j+1}, t_{j+1}) \frac{j}{3} \right] \quad j = 1,\, 2,\, 3 \tag{42}$$

$$\boldsymbol{\psi}_{i/\mathbf{x}_1} = \begin{bmatrix} 2(x_1 + \mu) & 2y_1 & 0 & 0 \\ v_{x,1} & v_{y,1} & (x_1 + \mu) & y_1 \end{bmatrix} \tag{43}$$

$$\boldsymbol{\psi}_{f/\mathbf{x}_4} = \begin{bmatrix} 2(x_4 + \mu - 1) & 2y_1 & 0 & 0 \\ v_{x,4} & v_{y,4} & (x_4 + \mu - 1) & y_4 \end{bmatrix} \tag{44}$$

Inequality constraints

$$\frac{\partial \mathbf{C}_{ms}}{\partial \mathbf{y}} = \begin{bmatrix} \boldsymbol{\eta}_{1/x,y} & \mathbf{0}_{2x4} & \mathbf{0}_{2x4} & \mathbf{0}_{2x4} & \mathbf{0}_{2x1} & \mathbf{0}_{2x1} \\ \mathbf{0}_{2x4} & \boldsymbol{\eta}_{1/x,y} & \mathbf{0}_{2x4} & \mathbf{0}_{2x4} & \mathbf{0}_{2x1} & \mathbf{0}_{2x1} \\ \mathbf{0}_{2x4} & \mathbf{0}_{2x4} & \boldsymbol{\eta}_{1/x,y} & \mathbf{0}_{2x4} & \mathbf{0}_{2x1} & \mathbf{0}_{2x1} \\ \mathbf{0}_{2x4} & \mathbf{0}_{2x4} & \mathbf{0}_{2x4} & \boldsymbol{\eta}_{1/x,y} & \mathbf{0}_{2x1} & \mathbf{0}_{2x1} \\ \mathbf{0}_{1x4} & \mathbf{0}_{1x4} & \mathbf{0}_{1x4} & \mathbf{0}_{1x4} & 1 & -1 \end{bmatrix} \tag{45}$$

$$\boldsymbol{\eta}_{j/x,y} = \begin{bmatrix} -2(x_j + \mu) & -2y_j & 0 & 0 \\ -2(x_j + \mu - 1) & -2y_j & 0 & 0 \end{bmatrix} \qquad j = 1,\, 2,\, 3,\, 4 \tag{46}$$

2.3.3 Solution of the problem

This problem too has been solved with the same procedure of subsubsection 2.2.3, using the initial guess in Table 8 for computing the mesh points.

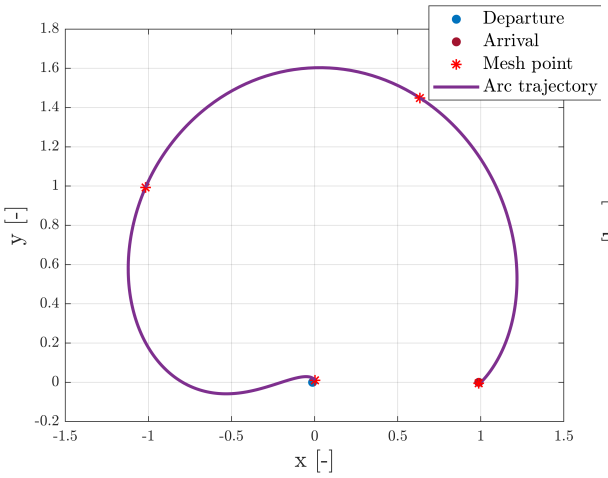


Figure 12: Earth-Moon transfer multiple shooting with gradients
@Earth-Moon Barycenter
Earth-Moon Adimensional rotating frame

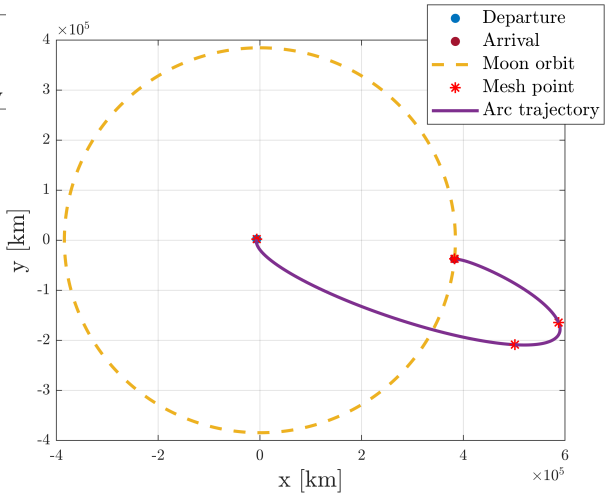


Figure 13: Earth-Moon transfer multiple shooting with gradients
@Earth
Earth Centered Inertial Frame

$r_{x,0}$ $[DU]$	$r_{y,0}$ $[DU]$	$v_{x,0}$ $[VU]$	$v_{y,0}$ $[VU]$	t_i $[TU]$	t_f $[TU]$
0.001697	0.009907	-6.224057	8.699482	2.194	6.191

Table 10: Multiple shooting solution in the Earth-Moon rotating frame.

2.4 Solutions comparison

For a solution to be accepted it is required that $\max\{||\boldsymbol{\psi}_i||, ||\boldsymbol{\psi}_f||\} < 10^{-8}$, as shown in Table 11 this condition has been fulfilled for all the three problems.
By comparing the solution found with simple shooting without providing the gradients and by providing it, it can be easily observed that in the presence of the gradients computed analytically has been improved a lot in the computational time, in fact, it drops down from approximately 8 seconds to 1 second, with the same accuracy in computing the Δv, and Δt.
In the multiple shooting due to the presence of a higher number of propagations and a higher number of constraints, the computational time increases even if the gradients are provided, but this leads to a more accurate solution in terms of Δv, and Δt, as shown in Table 11.

Problem	**Gradients**	Δt $[days]$	Δv $[km/s]$	CPU_{time} $[s]$	$\|\|\boldsymbol{\psi}_i\|\|$ $[-]$	$\|\|\boldsymbol{\psi}_f\|\|$ $[-]$
Single Shooting	False	17.391	4.106215	≈ 8	2.8703e-17	9.9129e-13
Single Shooting	True	17.391	4.106215	≈ 1	1.3878e-17	2.1603e-12
Multiple Shooting	True	17.358	4.104918	≈ 12	1.3878e-17	9.7117e-17

Table 11: Comparison between the simple shooting without and with gradients, and multiple shooting

2.5 N-body propagation

2.5.1 Identification of the initial epoch

Before proceeding with the N-body propagation it is necessary to identify the epoch in which the configuration of the Earth-Moon-Sun is similar to the one of the 2D problem, by comparing the angle between the rotating frame $x - axis$, aligned to the position vector from the Earth-Moon System Barycenter (EMB) to the Moon, and the Sun direction.
The computation of the angle for a specific epoch has been done by using the following algorithm:

- extraction of the Moon state, and Sun position, both with respect to EMB, from the kernels, by using `cspice_spkezr` and `cspice_spkpos` respectively;
- computation of the normal of the plane in which the Earth's and Moon's motion occurs, by the normalizing the cross product between the Moon position and Moon velocity obtained in the previous step;
- projection of the EMB-Sun vector onto the plane;
- computation of the angle.

The Figure 14 shows the evolution of the angle starting for a period of 40 days, starting from the `2024 Sep 28 00:00:00.000 TDB`. The figure shows a periodicity of approximately the period of the Moon around the Earth. The starting epoch has been found by solving a zero-finding

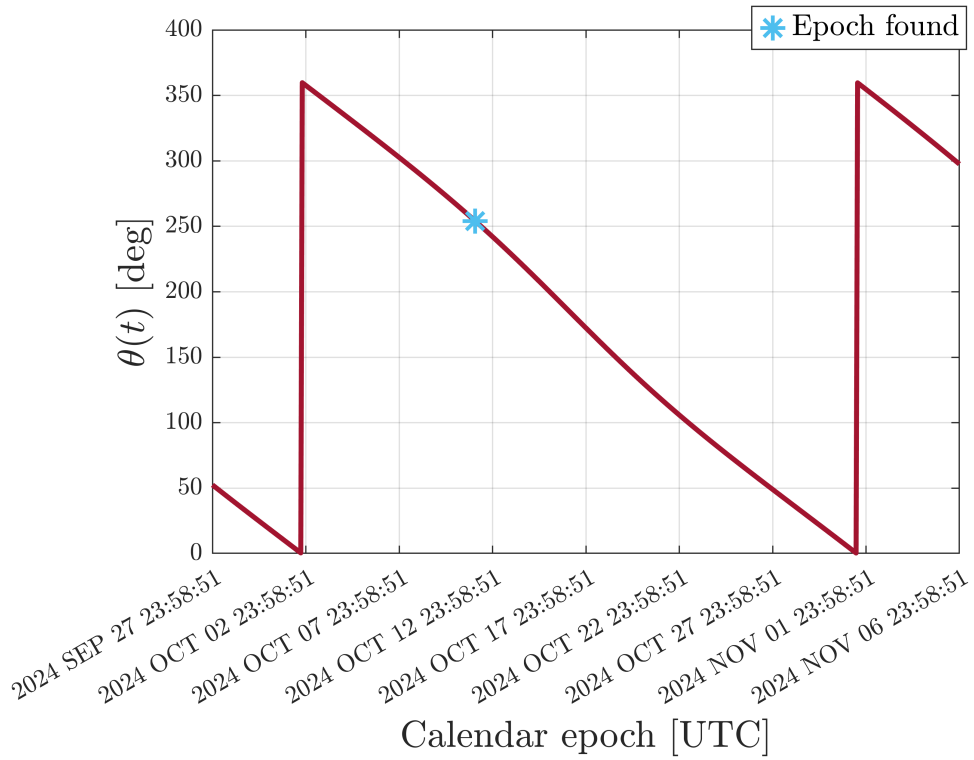


Figure 14: Evolution of Moon-EMB-Sun angle

problem by checking when the angle obtained from the algorithm is equal to the initial angle obtained from the initial condition. This problem has been solved using MATLAB `fsolve`, a crucial point for the convergence to a date after the `2024 Sep 28 00:00:00.000 TDB` is the initial guess. In this case any date between `2024 Oct 3 00:00:00.000 TDB` and `2024 Nov 1 00:00:00.000 TDB`, would be correct.

Symbol	**Calendar epoch** (UTC)
t_i	2024-10-12 01:38:29.947
t_f	2024-10-29 11:01:22.432

$r_{x,0}$ $[km]$	$r_{y,0}$ $[km]$	$r_{z,0}$ $[km]$
-5694.76331513	3226.21820362	0.0

$v_{x,0}$ $[km/s]$	$v_{y,0}$ $[km/s]$	$v_{z,0}$ $[km/s]$
-5.41087971	-9.55102146	0.0

Table 12: Initial epoch, final epoch, and initial state in Earth-centered inertial frame.

2.5.2 Propagation

After the initial and final epochs have been found, is possible to perform an n-body propagation, by propagating the initial state Table 12, by taking the position of the bodies from kernels. From results in Figure 15 and Table 13 it is possible to notice that the transfer has not been achieved, this happened because of all the simplification made in PBRFBP, and because typically this kind of analysis might be useful for, giving some preliminary estimation on the transfer cost if the transfer time is given, or for characterizing the solution structure by identifying families of transfer orbits. In the case of a real transfer an approach, in which the objective function and the constraints are computed using the real position of the Earth, Moon, and Sun would lead to a more accurate solution.

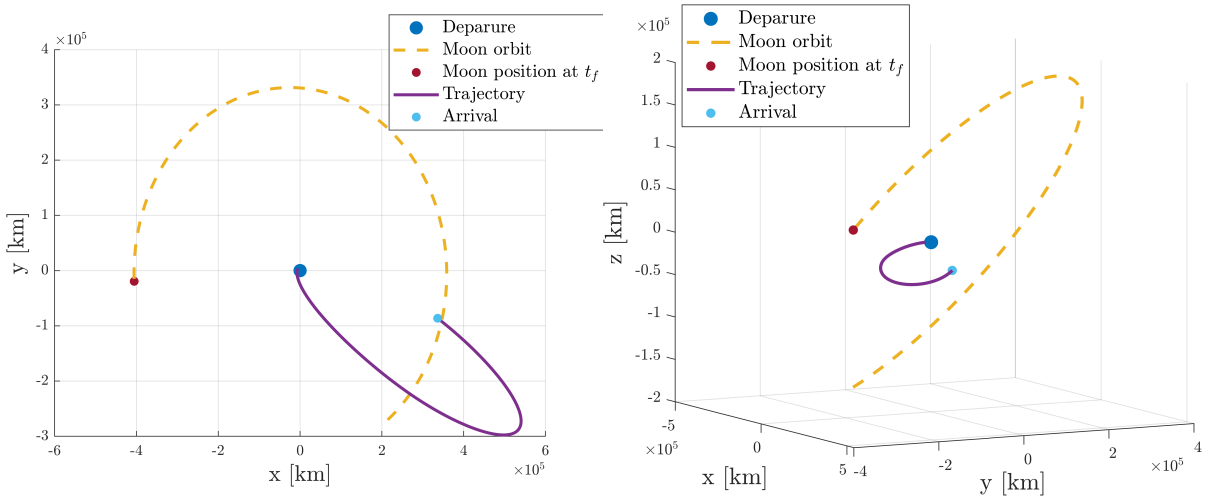


Figure 15: Earth-Moon transfer from two different point of view
@Earth Earth Centered Inertial Frame

Error	**Value**	**Units**
$\|\|\boldsymbol{r}(t_f) - \boldsymbol{r}_M(t_f)\|\|$	7.4497e+5	km
$\|\|\boldsymbol{v}(t_f) - \boldsymbol{v}_M(t_f)\|\|$	1.6961	km/s

Table 13: Final state error with respect to Moon state at final time t_f.

3 Continuous guidance

Exercise 3

A low-thrust option is being considered to perform an orbit raising maneuver using a low-thrust propulsion system in Earth orbit. The spacecraft is released on a circular orbit on the equatorial plane at an altitude of 800 km and has to reach an orbit inclined by 0.75 deg on the equatorial plane at 1000 km. This orbital regime is characterized by a large population of resident space objects and debris, whose spatial density q can be expressed as:

$$q(\rho) = \frac{k_1}{k_2 + \left(\frac{\rho - \rho_0}{DU}\right)^2}$$

where ρ is the distance from the Earth center. The objective is to design an optimal orbit raising that minimizes the risk of impact, that is to minimize the following objective function

$$F(t) = \int_{t_0}^{t_f} q(\rho(t))\,\mathrm{d}t.$$

The parameters and reference Distance Unit to be considered are provided in Table 14.

Symbol	Value	Units	Meaning
h_i	800	km	Altitude of departure orbit
h_f	1000	km	Altitude of arrival orbit
Δi	0.75	deg	Inclination change
R_e	6378.1366	km	Earth radius
μ	398600.435	$\mathrm{km}^3/\mathrm{s}^2$	Earth gravitational parameter
ρ_0	$750 + R_e$	km	Reference radius for debris flux
k_1	1×10^{-5}	DU^{-1}	Debris spatial density constant 1
k_2	1×10^{-4}	DU^2	Debris spatial density constant 2
m_0	1000	kg	Initial mass
$T_{\max}$	3.000	N	Maximum thrust
I_{sp}	3120	s	Specific impulse
DU	7178.1366	km	Distance Unit
MU	m_0	kg	Mass Unit

Table 14: Problem parameters and constants. The units of time TU and velocity VU can be computed imposing that the scaled gravitational parameter $\overline{\mu} = 1$.

1) Plot the debris spatial density $q(\rho) \in [h_i - 100;\, h_f + 100]$ km and compute the initial state and target orbital state, knowing that: i) the initial and final state are located on the x-axis of the equatorial J2000 reference frame; ii) the rotation of the angle Δi is performed around the x-axis of the equatorial J2000 reference frame (RAAN = 0).

2) Adimensionalize the problem using as reference length $DU = \rho_i = h_i + R_e$ and reference mass $MU = m_0$, imposing that $\mu = 1$. Report all the adimensionalized parameters.

3) Using the PMP, write down the spacecraft equations of motion, the costate dynamics, and the zero-finding problem for the unknowns $\{\boldsymbol{\lambda}_0, t_f\}$ with the appropriate transversality condition. **Hint**: the spacecraft has to reach the target state computed in point 1).

4) Solve the problem considering the data provided in Table 14. To obtain an initial guess for the costate, generate random numbers such that $\lambda_{0,i} \in [-250; +250]$ while $t_f \approx 20\pi$.

Report the obtained solution in terms of $\{\boldsymbol{\lambda}_0, t_f\}$ and the error with respect to the target. Assess your results exploiting the properties of the Hamiltonian in problems that are not time-dependent and time-optimal solution. Plot the evolution of the components of the primer vector $\boldsymbol{\alpha}$ in a NTW reference frame[‡].

5) Solve the problem for a lower thrust level $T_{\max} = 2.860$ N. Compare the new solution with the one obtained in the previous point. **Hint**: exploit numerical continuation.

(11 points)

3.1 Analysis of the problem

3.1.1 Debris spatial density

In order to successfully achieve the mission it is fundamental to comprehend the behavior of the debris spatial density within a designed range of altitude. From Figure 16 is possible to notice that the behavior of $q(\rho)$ is similar to a Cauchy distribution. The function has a peak in correspondence of $\rho_0 = 7128.1366\,km$ then it decreases on both sides. In the figure are also represented the initial and final distance ρ_i, ρ_f respectively, it is possible to observe that the S/C goes from a region with a higher density of debris to a lower one.

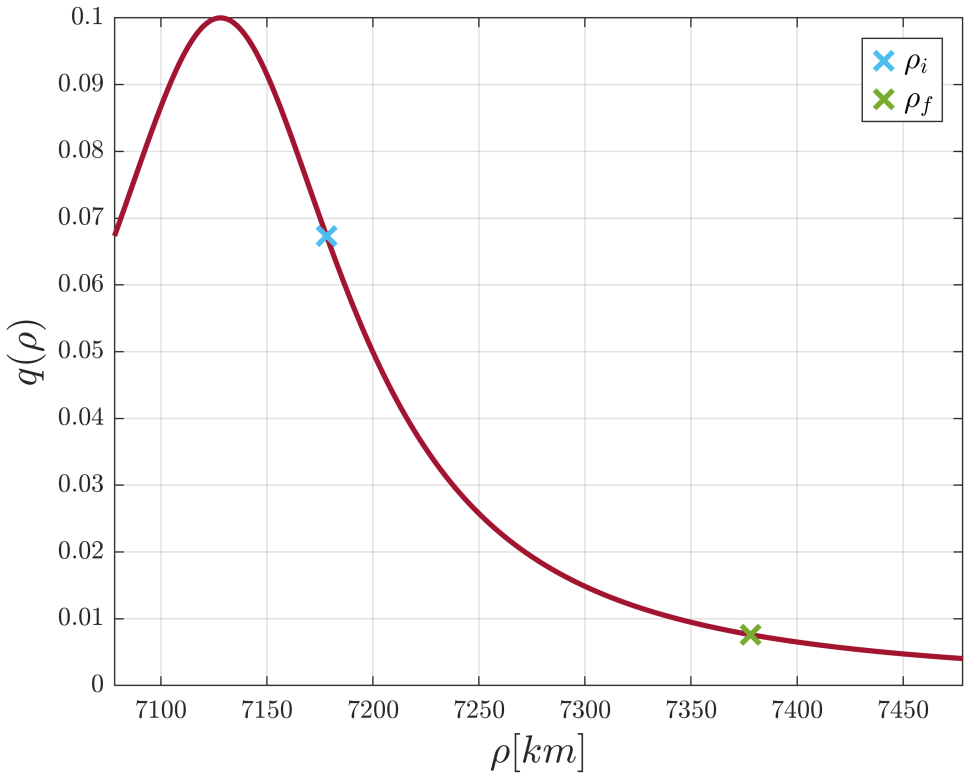


Figure 16: Debris spatial density, for a range of altitude of $[h_i - 100;\, hf + 100]\,km$

3.1.2 Initial state and final state

The correct computation of the initial and final state is fundamental for the statement of the problem.
Considering that both the initial and final state lies on $x-axis$ of the equatorial J2000 reference

[‡]The T-axis is aligned with the velocity, the N-axis is normal to the angular momentum, while the W-axis is pointing inwards, i.e., towards the Earth.

frame, the y and z components of position for both the states are equal to zero.
Given that both orbits are circular the only velocity component different from zero in the initial state is v_y, computed through Equation 47.

$$v_{y,i} = \sqrt{\frac{\mu}{R_e + h_i}} \tag{47}$$

For the final state, also v_z is different from zero due to the change in the orbit inclination. The components are computed using Equation 48.

$$v_{y,f} = \sqrt{\frac{\mu}{R_e + h_f}} cos(\Delta i) \qquad v_{z,f} = \sqrt{\frac{\mu}{R_e + h_f}} sin(\Delta i) \tag{48}$$

$r_{x,i}$ $[km]$	$r_{y,i}$ $[km]$	$r_{z,i}$ $[km]$	$v_{x,i}$ $[km/s]$	$v_{y,i}$ $[km/s]$	$v_{z,i}$ $[km/s]$
7178.136600	0.0	0.0	0.0	7.45183148	0.0

$r_{x,f}$ $[km]$	$r_{y,f}$ $[km]$	$r_{z,f}$ $[km]$	$v_{x,f}$ $[km/s]$	$v_{y,f}$ $[km/s]$	$v_{z,f}$ $[km/s]$
7378.136600	0.0	0.0	0.0	7.34950906	0.09621034

Table 15: Initial and target state in Earth-centered equatorial J2000 inertial frame.

3.2 Adimensionalizzation

To achieve a faster convergence to the optimal solution the problem has been adimensionalized. Imposing the scaled gravitational parameter $\overline{\mu} = 1$, and using the distance unit DU, the adimensionalized time TU has been computed.

$$TU = \sqrt{\frac{DU^3}{\mu}} \tag{49}$$

Using TU, DU, and MU, reported in Table 14, the unit values for velocity and the force, VU, and FU respectively, has been computed.

TU $[s]$	VU $[km/s]$	FU $[N]$
963.271462	7.451831	7735.962056

3.3 Formulation of the problem

The problem under analysis is a continuous guidance problem, whose general statement is reported in Equation 50.

$$\min_{\boldsymbol{u} \in \Omega} J(\boldsymbol{u}) := \int_{t_0}^{t_f} l(\mathbf{x}, \boldsymbol{u}, t)\, dt \quad s.t \begin{cases} \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \boldsymbol{u}, t) \\ \mathbf{x}(t_0) = \mathbf{x}_0 \\ \mathbf{x}_i(t_f) = \mathbf{x}_{f,i} \quad i = 1, ..., I < n \end{cases} \tag{50}$$

The equations of motion are the following:

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v} \\ \dot{\boldsymbol{v}} = -\frac{\mu}{r^3}\boldsymbol{r} + u(m,\, \lambda_v,\, \lambda_m)\frac{T_{max}}{m}\hat{\boldsymbol{\alpha}} \\ \dot{m} = -u(m,\, \lambda_v,\, \lambda_m)\frac{T_{max}}{I_{sp} g_0} \end{cases} \tag{51}$$

The boundary constraints are:

$$\begin{cases} \mathbf{x}(t_0) &= \mathbf{x}_i \\ m(t_0) &= m_0 \\ \mathbf{x}(t_f) &= \mathbf{x}_f \\ \lambda_m(t_f) &= 0 \end{cases} \tag{52}$$

The objective function is:

$$l(\mathbf{x}, \boldsymbol{u}, t) = q(\rho) = \frac{k_1}{k_2 + \left(\dfrac{\rho - \rho_0}{DU}\right)^2} \tag{53}$$

The application of the Pontryagin Maximum Principle (PMP) is used to find the best possible control taking a dynamical system from an initial state to a final state. In the PMP the optimal control u is determined at each instant to render the Hamiltonian, Equation 54 a minimum over all admissible control functions.

$$H = l(\mathbf{x}, \boldsymbol{u}, t) + \boldsymbol{\lambda} \cdot \mathbf{f}(\mathbf{x}, \boldsymbol{u}, t) \tag{54}$$

PMP also implies that the control action must belong to a set of admissible controls defined in Equation 55

$$\Omega = \{(u, \hat{\boldsymbol{\alpha}}) \quad s.t. \quad u \in [0,\, 1], \quad \hat{\alpha} = 1\} \tag{55}$$

The optimal thrust-pointing direction is Lawden's primer vector, Equation 56.

$$\hat{\boldsymbol{\alpha}} = -\frac{\boldsymbol{\lambda_v}}{\lambda_v} \tag{56}$$

By substituting the previous assumption in Equation 54, this becomes:

$$H = \frac{k_1}{k_2 + \left(\dfrac{\rho - \rho_0}{DU}\right)^2} + \boldsymbol{\lambda_r} \cdot \boldsymbol{v} - \frac{\mu}{r^3}(\boldsymbol{r} \cdot \boldsymbol{\lambda_v}) + u(m,\, \lambda_v,\, \lambda_m)\frac{T_{max}}{I_{sp} g_0}\left(-\frac{\lambda_v}{m} I_{sp} g_0 - \lambda_m\right) \tag{57}$$

This result shows that the Optimal Control Problem is solved for $u = 1$. The TPBVP is composed by Equation 51 and by Equation 58, which represent the costate dynamics.

$$\dot{\boldsymbol{\lambda}} = -\frac{\partial H}{\partial \mathbf{x}} = \begin{cases} \dot{\boldsymbol{\lambda}}_{\boldsymbol{r}} = \dfrac{2k_1(\rho - \rho_0)}{\left(k2 + \left(\dfrac{\rho - \rho_0}{DU}\right)^2\right)^2 \rho DU^2}\boldsymbol{r} - \dfrac{3\mu}{r^5}(\boldsymbol{r} \cdot \boldsymbol{\lambda_v})\boldsymbol{r} + \dfrac{\mu}{r^3}\boldsymbol{\lambda_v} \\ \dot{\boldsymbol{\lambda}}_{\boldsymbol{v}} = -\boldsymbol{\lambda_r} \\ \dot{\lambda}_m = -u(m,\, \lambda_v,\, \lambda_m)\dfrac{\lambda_v T_{max}}{m^2} \end{cases} \tag{58}$$

The problem can be seen as a zero-finding problem, so it has been solved by employing MATLAB solver `fsolve`. The quantities that need to be zero are the residuals, expressed in Equation 59, where the first two conditions are known, the third is used because there is not a known final value for the mass at t_f, and the last one is the transversality condition, which is introduced by the Hamiltonian because the final time is a free variable, and requires solving for t_f.

$$\begin{cases} \boldsymbol{r}(t_f) - \boldsymbol{r}_f = 0 \\ \boldsymbol{v}(t_f) - \boldsymbol{v}_f = 0 \\ \lambda_m(t_f) = 0 \\ H(t_f) = 0 \end{cases} \tag{59}$$

The problem consists in find $\{\boldsymbol{\lambda}_0,\, t_f\}$ such that the integration of the Equation 51 and Equation 58 produce the solutions $\mathbf{x}(t)$ and $\boldsymbol{\lambda}(t)$ such that Equation 59 are satisfied.

3.4 Solution for $T_{max} = 3.000$N

To solve the problem eight parameters are needed, the initial guess of the costate and the adimensionalized final time, $\{\boldsymbol{\lambda}_0, t_f\}$. These parameters are used in `fsolve` which works in the following way:

- $\boldsymbol{\lambda}_0$, $\mathbf{x}_i$ are integrated through Equation 51 and Equation 58;
- extraction of state and costate at t_f;
- computation of the residuals through Equation 59.

This procedure is repeated iteratively until the residuals go under a settled tolerance.

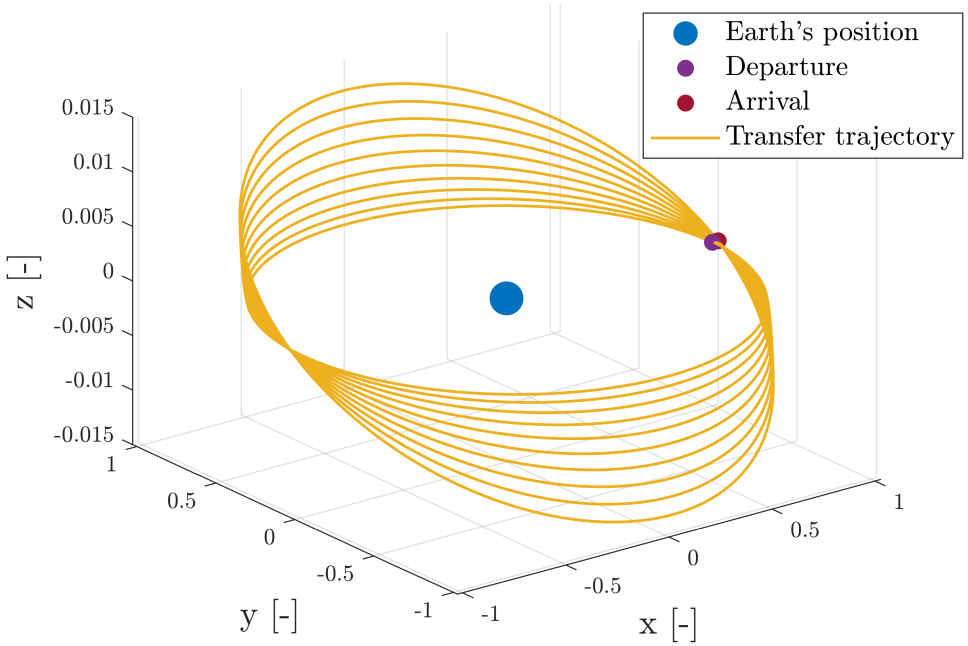


Figure 17: Orbit raising maneuver for $T_{max} = 3.000\,N$ @Earth Earth-Centered Equatorial J2000 Inertial Adimensionalized Frame

t_f [mins]	m_f $[kg]$
1035.1974	993.9120

λ_{0,r_x}	λ_{0,r_y}	λ_{0,r_z}	λ_{0,v_x}	λ_{0,v_y}	λ_{0,v_z}	$\lambda_{0,m}$
-214.9812	-10.3659	0.8856	-10.3929	-214.6104	-112.9453	2.5964

Table 16: Optimal orbit raising transfer solution ($T_{\max} = 3.000$ N).

Error	**Value**	**Units**
$\|\|\mathbf{r}(t_f) - \mathbf{r}_f\|\|$	1.9512e-07	km
$\|\|\mathbf{v}(t_f) - \mathbf{v}_f\|\|$	2.1433e-07	m/s

Table 17: Final state error with respect to target position and velocity ($T_{\max} = 3.000$ N).

An important check to verify the validity of the results involves analyzing the Hamiltonian. According to theory, if the Hamiltonian (H) is not an explicit function of time, it remains constant. Figure 18 illustrates the results: considering the small variations due to the numerical precision of the computation, it can be concluded that the hypothesis of a constant H is satisfied.

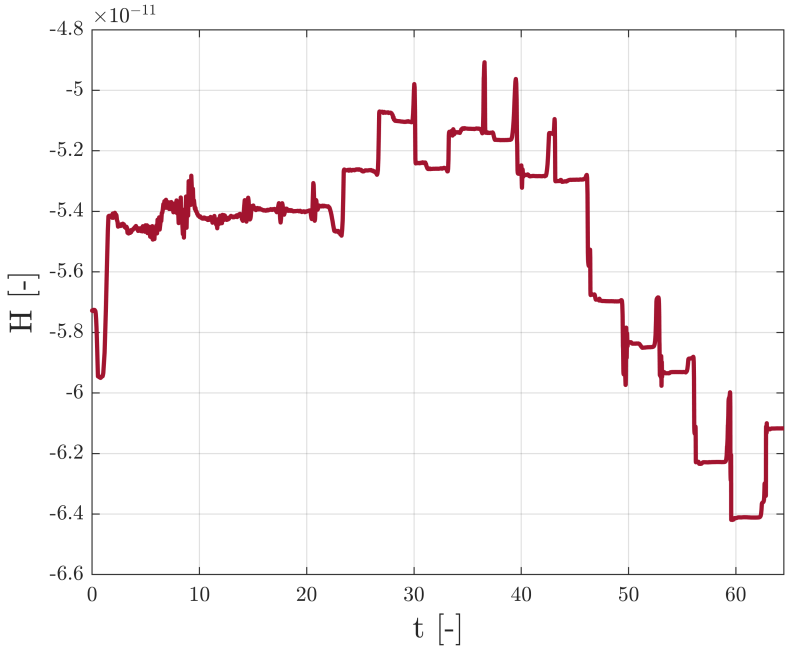


Figure 18: Hamiltonian

The computation of Lawden's primer vector, ($\hat{\boldsymbol{\alpha}}$), components in the NTW reference frame has been done by propagating the initial state and the initial costate until the final time, then for each time instant $\hat{\boldsymbol{\alpha}}$ has been computed using Equation 56, then the rotation matrix needs to be computed, Equation 60 and after the vector is rotated, Equation 61.

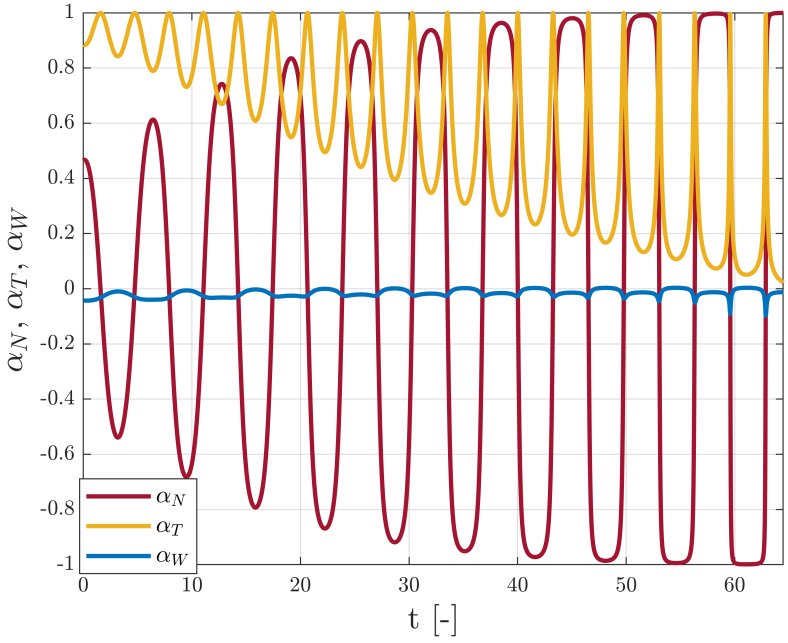


Figure 19: Components of the Lawden's primer vector in the NTW reference frame

$$\mathbf{T} = \frac{\boldsymbol{v}}{v}, \quad \boldsymbol{W} = -\frac{\boldsymbol{r}}{r}, \quad \boldsymbol{N} = \frac{\boldsymbol{T} \times \boldsymbol{N}}{||\boldsymbol{T} \times \boldsymbol{N}||} \qquad \mathbf{R}_{NTW} = \begin{bmatrix} \boldsymbol{N} & \boldsymbol{T} & \boldsymbol{W} \end{bmatrix}^T \tag{60}$$

$$\hat{\boldsymbol{\alpha}}_{NTW} = R_{NTW} \hat{\boldsymbol{\alpha}} \tag{61}$$

The outcomes showed in Figure 19 are in line with what one could expect; in fact, the N component shows positive peaks when the S/C crosses the ascending node, while negative peaks

when it crosses the descending node. At the final time, the T and W components approach zero while the N is maximum.

3.5 Solution for $T_{max} = 2.860$N

In the same way, the problem has been solved for a lower value of thrust. This has been done by exploiting numerical continuation on the guessed costate and final time. By using the solution found previously, the problem has been solved for a new low level of thrust. This procedure continues iteratively until the value of thrust reaches $T_{max} = 2.860\,N$.
The numerical continuation reduces significantly the computational time and solutions are still found. By comparing the results of the two solutions in terms of time and final mass, Table 16

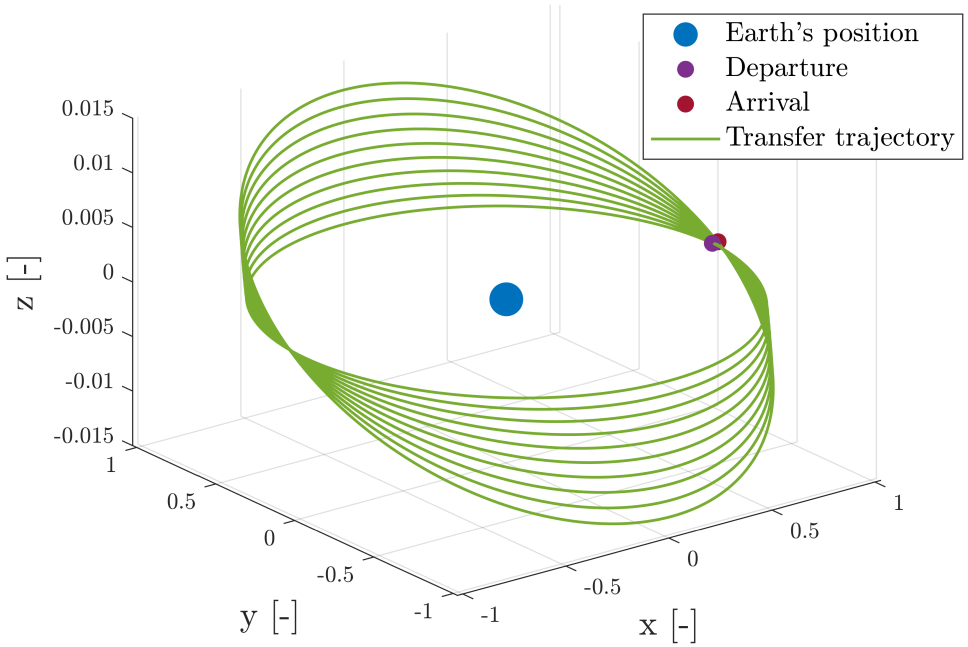


Figure 20: Orbit raising maneuver for $T_{max} = 2.860\,N$ @Earth Earth-Centered Equatorial J2000 Inertial Adimensionalized Frame

and Table 18 it is evident how the transfer achieved with $T_{max} = 2.860\,N$ allows a higher dry mass, $\approx 0.31\,kg$, and also a lower Δt of ≈ 4 mins.

t_f [mins]	m_f $[kg]$
1030.9760	994.2198

λ_{0,r_x}	λ_{0,r_y}	λ_{0,r_z}	λ_{0,v_x}	λ_{0,v_y}	λ_{0,v_z}	$\lambda_{0,m}$
-593.1529	-11.5904	2.2973	-11.9293	-592.7890	-920.3698	17.3814

Table 18: Optimal orbit raising transfer solution ($T_{\max} = 2.860$ N).

Error	**Value**	**Units**
$\|\|\mathbf{r}(t_f) - \mathbf{r}_f\|\|$	8.7117e-08	km
$\|\|\mathbf{v}(t_f) - \mathbf{v}_f\|\|$	9.1831e-08	m/s

Table 19: Final state error with respect to target position and velocity ($T_{\max} = 2.860$ N).